Integrated Software Project

Assignment 2

Formal System Proposal and Specification

Assessment information	Details
Module code	ISJ107V
Assessment weighting	10% of the final module mark
Total marks	50 marks
Due date	19 September 2026
Assessment type	Individual
Submission	Through EC

Instructions

- Answer all five questions in the order presented and retain the question number
- Use realistic emerging technologies, such as AI or machine learning, blockchain, IoT, AR or VR, edge computing, or another lecturer-approved technology.
- Write requirements as clear, testable statements and assign unique identifiers such as FR-01 and NFR-01.
- Use a recognised UML tool. Every diagram must be readable, labelled and consistent with the written requirements.

Question 1 Total Marks [8]

Introduce the proposed project and establish why it should be developed.

1.1 Provide the project title and a concise description of the proposed software system, its intended users and the emerging technology to be integrated. (2)

1.2 Formulate a clear problem statement that explains the current problem, who is affected and the consequences of leaving it unresolved. (3)

1.3 State one project aim, three measurable objectives, the main scope boundaries and two success criteria. (3)

Question 2 Total Marks [15]

Specify what the proposed system must do and the quality standards it must satisfy.

2.1 Develop ten prioritized functional requirements. Use unique identifiers and the format "The system shall ...". Include acceptance criteria for each requirement. (10)

2.2 Develop four measurable non-functional requirements using relevant FURPS+ categories: functionality, usability, reliability, performance, supportability and applicable constraints. (2)

2.3 Specify three testable requirements that apply directly to the selected emerging technology, such as model accuracy, inference latency, sensor reliability, transaction integrity or device compatibility. (3)

Recommended requirement format:

ID	Priority	Requirement	Acceptance criterion
FR-01	Must	The system shall allow an authorised user to ...	Given ... when ... then ...
NFR-01	Must	The system shall return ... within 3 seconds ...	Measured response time is 3 seconds or less.

Question 3 Total Marks [10]

Define and justify the technical environment required to develop, test and operate the system.

3.1 Present the proposed technology stack, including the user interface, application layer, database or dataset, APIs, development tools and deployment platform. Briefly justify the main choices. (3)

3.2 Provide a minimum and recommended technical hardware requirements table covering processor, memory, storage, network or connectivity, client devices and any specialised hardware required by emerging technology. (4)

3.3 Explain the proposed architecture and deployment approach, including how major components communicate and how scalability, security, backup and recovery will be addressed. (3)

Question 4 Total Marks [13]

Create the following UML models. Use the same actor, class, component and requirement names throughout the submission.

4.1 Use-case diagram: show the system boundary, primary and supporting actors, major use cases, and appropriate include, extend or generalisation relationships where relevant. (4)

4.2 Activity diagram: model one important end-to-end workflow, including start and end nodes, decisions, alternative paths and the emerging-technology step. (3)

4.3 Class diagram: show the main domain classes, important attributes and operations, associations, multiplicities and inheritance or composition where applicable. (3)

4.4 Sequence diagram: show the time-ordered interaction for one core use case across the user interface, application logic, data service and emerging-technology component. (3)

Question 5 Total Marks [13]

Explain how the proposed design will be implemented and evaluated.

5.1 Describe the database or dataset and the API or interface through which data will be exchanged. Identify the main data entities, inputs, outputs, validation and privacy requirements. (2)

5.2 Explain where and how the emerging-technology component will be integrated into the software workflow. Include one short pseudocode fragment or interface contract. (2)

5.3 Provide a concise testing and evaluation plan covering functional testing, one measurable emerging-technology metric, user acceptance testing and one important technical risk with mitigation. (2)